

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** tert-Butyl methyl ether

**CAS No** 1634-04-4

**Synonyms** 2-Methyl-2-methoxy propane; MTBE; Methyl tert-butyl ether

**Product Code** E127-4; XXE127RS200; NC1240503; XXE127U200LI; NC1568440; E127RS1350ASME; NC1561779

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** CHEMTREC®  
03 9757 4559 or +613 9757 4559

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Skin Corrosion/Irritation

Category 2

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor

H315 - Causes skin irritation

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P312 - Call a POISON CENTER or doctor if you feel unwell

P363 - Wash contaminated clothing before reuse

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Contains a known or suspected endocrine disruptor

Contains a substance on the National Authorities Endocrine Disruptor Lists

**Section 3 - Composition and Information on Ingredients**

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| Methyl tert-butyl ether | 1634-04-4 | >95      |

**Section 4 - First Aid Measures****Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | medical attention.                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting     |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                  |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Ensure adequate ventilation.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges. Use only under a chemical fume hood. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

**Conditions for Safe Storage, Including any Incompatibilities**

Flammables area. Keep away from heat, sparks and flame. Keep container tightly closed in a dry and well-ventilated place. May form explosive peroxides on prolonged storage.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component               | Australia                                                                               | New Zealand WEL                                                                         | ACGIH TLV   | The United Kingdom                                                                                                  | Germany                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl tert-butyl ether | STEL: 75 ppm<br>STEL: 275 mg/m <sup>3</sup><br>TWA: 25 ppm<br>TWA: 92 mg/m <sup>3</sup> | TWA: 25 ppm<br>TWA: 92 mg/m <sup>3</sup><br>STEL: 75 ppm<br>STEL: 275 mg/m <sup>3</sup> | TWA: 50 ppm | STEL: 100 ppm 15 min<br>STEL: 367 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 183.5 mg/m <sup>3</sup> 8 hr | TWA: 50 ppm (8 Stunden). AGW - exposure factor 1.5<br>TWA: 180 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1.5<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 180 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 75 ppm<br>Höhepunkt: 270 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                                                            |
|----------------|-------------------|-----------------|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| Nitrile rubber | < 211 minutes     | 0.38 mm         | AS/NZS 2161     | Permeation rate 1 µg/cm <sup>2</sup> /min                                                                                 |
| Viton (R)      | < 152 minutes     | 0.7 mm          |                 | Permeation rate 17 µg/cm <sup>2</sup> /min As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene       |                   |                 |                 |                                                                                                                           |
| Natural rubber |                   |                 |                 |                                                                                                                           |
| PVC            |                   |                 |                 |                                                                                                                           |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                                |                                             |
|------------------------------------------------|------------------------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Colorless                                      |                                             |
| <b>Physical State</b>                          | Liquid                                         |                                             |
| <b>Odor</b>                                    | Petroleum distillates                          |                                             |
| <b>Odor Threshold</b>                          | No data available                              |                                             |
| <b>pH</b>                                      | No information available                       |                                             |
| <b>Melting Point/Range</b>                     | -110 °C / -166 °F                              |                                             |
| <b>Softening Point</b>                         | No data available                              |                                             |
| <b>Boiling Point/Range</b>                     | 54 - 56 °C / 129.2 - 132.8 °F                  |                                             |
| <b>Flash Point</b>                             | -28 °C / -18.4 °F                              | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available                              |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                                 | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.6 vol%<br><b>Upper</b> 8.4 vol% |                                             |
| <b>Vapor Pressure</b>                          | 268 mbar @ 20 °C                               |                                             |
| <b>Vapor Density</b>                           | 0.2                                            | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.740                                          |                                             |
| <b>Bulk Density</b>                            | Not applicable                                 | Liquid                                      |
| <b>Water Solubility</b>                        | 51 g/L (20°C)                                  |                                             |
| <b>Solubility in other solvents</b>            | No information available                       |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                 |                                             |
| Methyl tert-butyl ether                        | 1.06                                           |                                             |
| <b>Autoignition Temperature</b>                | 224 °C / 435.2 °F                              |                                             |
| <b>Decomposition Temperature</b>               | No data available                              |                                             |
| <b>Viscosity</b>                               | 0.36 mPa.s at 20 °C                            |                                             |
| <b>Explosive Properties</b>                    |                                                | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available                       |                                             |
| <b>Other information</b>                       |                                                |                                             |
| <b>Molecular Formula</b>                       | C5 H12 O                                       |                                             |
| <b>Molecular Weight</b>                        | 88.15                                          |                                             |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                            |
| <b>Stability</b>                        | Stable under normal conditions.                                                                       |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents.                                                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                              |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component               | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation            |
|-------------------------|---------------------------|-------------------------------|----------------------------|
| Methyl tert-butyl ether | LD50 = 2963 mg/kg ( Rat ) | LD50 = 10000 mg/kg ( Rabbit ) | LC50 = 85 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Based on available data, the classification criteria are not met

##### (d) respiratory or skin sensitization;

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

Mutagenic effects have occurred in experimental animals

##### (f) carcinogenicity;

Based on available data, the classification criteria are not met

The table below indicates whether each agency has listed any ingredient as a carcinogen  
Limited evidence of a carcinogenic effect

##### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

##### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

##### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

**Target Organs**

None known.

##### (j) aspiration hazard;

Based on available data, the classification criteria are not met

#### Other Adverse Effects

Tumorigenic effects have been reported in experimental animals.

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

### Ecotoxicity effects

Do not empty into drains. .

| Component               | Freshwater Fish                                                                      | Water Flea                               | Freshwater Algae                             | Microtox                                                                     |
|-------------------------|--------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------|
| Methyl tert-butyl ether | 887 mg/L LC50 96 h<br>100 mg/L LC50 96 h<br>929 mg/L LC50 96 h<br>672 mg/L LC50 96 h | EC50: = 542 mg/L, 48h<br>(Daphnia magna) | 800 mg/L EC50 > 72 h<br>184 mg/L EC50 = 96 h | EC50 = 11.4 mg/L 30 min<br>EC50 = 8.23 mg/L 5 min<br>EC50 = 9.67 mg/L 15 min |

### Persistence and Degradability

#### Persistence

Persistence is unlikely, based on information available.

#### Bioaccumulative Potential

Bioaccumulation is unlikely

| Component               | log Pow | Bioconcentration factor (BCF) |
|-------------------------|---------|-------------------------------|
| Methyl tert-butyl ether | 1.06    | No data available             |

### Mobility

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

### Endocrine Disruptor Information

| Component               | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Methyl tert-butyl ether | Group III Chemical                       |                                                  |                                         |

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

### Other Information

Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

### IMDG/IMO

|                      |                         |
|----------------------|-------------------------|
| UN-No                | UN2398                  |
| Proper Shipping Name | METHYL tert-BUTYL ETHER |
| Hazard Class         | 3                       |
| Packing Group        | II                      |

### ADG

|                      |                         |
|----------------------|-------------------------|
| UN-No                | UN2398                  |
| Proper Shipping Name | METHYL tert-BUTYL ETHER |
| Hazard Class         | 3                       |
| Packing Group        | II                      |

| Component                                    | Hazchem Code |
|----------------------------------------------|--------------|
| Methyl tert-butyl ether<br>1634-04-4 ( >95 ) | 3YE          |

**IATA**

**UN-No** UN2398  
**Proper Shipping Name** METHYL tert-BUTYL ETHER  
**Hazard Class** 3  
**Packing Group** II

**Environmental hazards** No hazards identified

**Special Precautions** No special precautions required

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                           | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-------------------------------------|-------------------------------------------------------------|------------------------|
| Methyl tert-butyl ether - 1634-04-4 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### **International Inventories**

| Component | AICS | NZIoC | EINECS | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL |
|-----------|------|-------|--------|--------|------|-----|------|-------|------|------|-------|------|
|-----------|------|-------|--------|--------|------|-----|------|-------|------|------|-------|------|



|                         |   |   |           |   |   |   |   |   |   |   |   |          |
|-------------------------|---|---|-----------|---|---|---|---|---|---|---|---|----------|
| Methyl tert-butyl ether | X | X | 216-653-1 | - | X | X | - | X | X | X | X | KE-23648 |
|-------------------------|---|---|-----------|---|---|---|---|---|---|---|---|----------|

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component                           | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|-------------------------------------|------------------------------------|------------------------------------------------------------------------|
| Methyl tert-butyl ether - 1634-04-4 | Annex I - Y40                      | Y40                                                                    |

| Component               | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyl tert-butyl ether | 1634-04-4 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

### Authorisation/Restrictions according to EU REACH

| Component               | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl tert-butyl ether | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**Predicted No Effect Concentration (PNEC)**  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail  
**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%

**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date** 21-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**